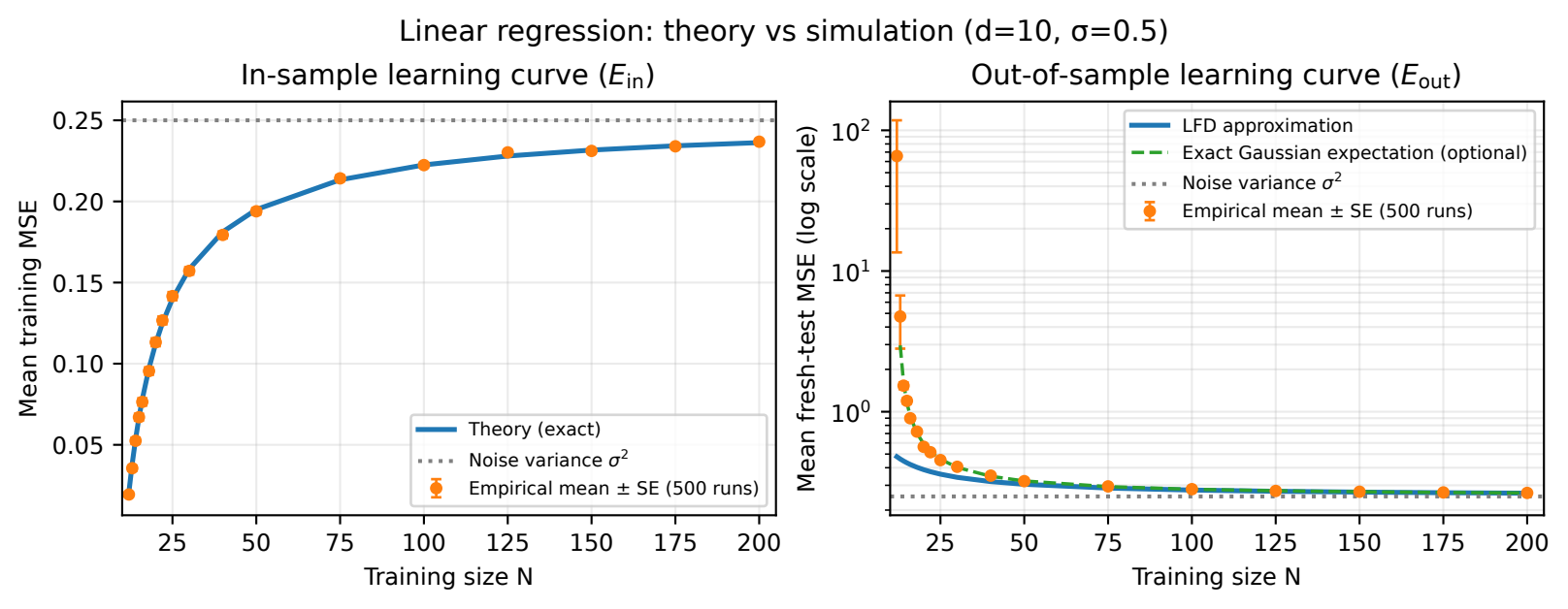
Linear regression: theory vs simulation (d=10, σ=0.5)
In-sample learning curve ($E_{in}$)
Mean training MSE
0.25
0.20
0.15
0.10
0.05
25
50
75
100
125
150
175
200
Training size N
Theory (exact)
Noise variance $\sigma^2$
Empirical mean ± SE (500 runs)
Out-of-sample learning curve ($E_{out}$)
Mean fresh-test MSE (log scale)
$10^2$
$10^1$
$10^0$
LFD approximation
Exact Gaussian expectation (optional)
Noise variance $\sigma^2$
Empirical mean ± SE (500 runs)
Training size N